PHILIPP AGRIC SCIENTIST
Vol. 93 No. 2, 224-231
June 2010

ISSN 0031-7454

Comparative Analysis of Energy Efficiency in Organic and Conventional Farming Systems: A Case Study of Black Carrot (*Daucus carota* L.) Production in Turkey

Yusuf Çelik*, **Kenan Peker** and **Cennet Oguz**

University of Selcuk, Department of Agricultural Economics, Faculty of Agriculture, 42079, Konya, Turkey
*Author for correspondence; e-mail: yucelik@hotmail.com; Tel.: + 90 332 2232870; Fax: + 90 332 2410108

This study compared the energy efficiency and cost/benefit ratio of conventional and organic farming systems for black carrot (*Daucus carota* L.) in Konya province, Turkey. In the research, energy consumption for different farm operations from land preparation to transportation was considered. The farms were selected through stratified random sampling. Data were collected through field survey of 56 respondents in carrot farms during production periods in 2007–2008. Black carrot production was considered for the study as it is a very good source of colored food. Therefore, black carrot production must take into consideration the requirements for eco-design requirements and energy labeling.

Based on the results, increasing amount of inputs was more effectively used in the organic farming system compared with the conventional system. Total energy requirement under conventional farming was 75,335.72 MJ ha^{-1} and 37,758.82 MJ ha^{-1} for organic farming. There was higher demand for energy in conventional farming (49.88%) than organic farming. On the other hand, the need for labor per hectare was higher for organic farming than for the conventional farming system. The energy output-input ratios for conventional and organic farming systems were 1.30 and 1.90, respectively. Moreover, the benefit-cost ratios were 1.83 and 2.05, respectively. Energy efficiency in the organic systems may increase by 1.32 kg ha^{-1} and by 0.23 kg ha^{-1} in the conventional systems.

Key Words: benefit-cost ratio, black carrot, energy efficiency, farming systems, labor

INTRODUCTION

Energy is an input used for various purposes such as increasing productivity, enhancing food security and contributing to rural economic development in the agricultural sector. The use of fossil energy and the concomitant emission of CO_2 in the agricultural operational sector – i.e., the use of farm machinery, irrigation, fertilization and chemical pesticides – amounts to merely 3.9% of the commercial energy use in parts of the world (Vlek et al. 2004). Also, energy input-output relationships in cropping systems vary with crops grown in sequence, type of soils, nature of tillage operations for seedbed preparation, the nature and the amount of organic manure, the chemical fertilizer used, plant protection measures, harvesting and, finally, yield levels.

Modern agriculture is generally applied in the developed countries, and requires more energy input compared with traditional agriculture (Turyereeba 2001). Turkey's agriculture also requires more energy consumption due to the increasing level of technology (Sayin et al. 2005). Producers currently use more energy to increase the total output because there is no chance to expand the size of arable lands and producers also do not have enough knowledge on energy efficiency. Agronomists have proposed several solutions such as integrated arable farming systems and low input or organic farming for the implementation of energy efficiency in agriculture as a

concept of sustainability (Vereijken 1997; Edwards 1987; Klimekova and Lehocka 2007). On the other hand, crop management systems need to be designed to help farmers in maintaining economic profitability, conserving external energy resources and, at the same time, becoming aware of their environmental responsibility (Pervanchon et al. 2002; Mader et al. 2002; Oguz et al. 2005). Thus, organic farming was developed by farmers who were concerned with both resource utilization and environment friendly production. Organic agriculture can reduce reliance on agro-chemical inputs as well as create environment friendly and economically sound agriculture (Pimentel et al. 2005).

A comparison of energy use in organic and non-organic production systems is crucial to understand the energy inefficiency of different farming systems and their potential for reducing energy consumption. Many studies have been conducted on energy input-output issues and economic analysis for various crops (Pretty 1995; Nguyen and Haynes 1995; Jianbo 2006; Mäder et al. 2002; Gündogmus and Bayramoglu 2005; Gundogmus 2006; Klimekova and Lehocka 2007). For farming systems to succeed, the **5E**s (**e**nergy efficiency, **e**conomic profitability, **e**nvironment friendly production, **e**xperiences or educational level of farmers and **e**thics of a profession) have to be taken into consideration in integrated farming system management (IFSM) (Farnworth et al. 2008; Tiwari et al. 1999).

This study comparatively analyzed the organic and conventional farming systems for the improvement of energy efficiency in the production of black carrot. Black carrot is a source of fresh vegetable produce, carrot juice and natural or organic pigment for food color. Therefore, black carrot production must take into consideration the requirements for eco-design and energy labeling. Organic and conventional farming systems were compared in terms of energy input, energy profit and benefit/cost for black carrot in order to increase energy productivity.

MATERIALS AND METHODS

The materials were collected, as energy use and input costs for conventional and organic black carrot production, in Konya Province, Turkey. The region is located in central Anatolia of Turkey (36° 41′ and 39° 16′ north latitude and 31° 14′ and 34° 26′ east longitude). Konya has a land area of 4,169,400 ha, 63.8% of which constitutes the farming area, equivalent to 2,659,890 ha. The annual average temperature in the region is approximately 11.3 °C while the annual total rainfall is 303.1 mm, 27.3% of which occurs from April to June (Anonymous 2007a). Konya province was chosen as sample area because it is the major black carrot producing area in Turkey. The Ministry of Agriculture and Rural Affairs (Konya branch) estimated that annual black carrot production was about 25,000 tons in Konya province where 90% of the black carrot produce in Turkey is harvested (Anonymous 2007b). The data were collected from a survey of 56 black carrot farms in eight villages. These farms which produce carrot using both organic and conventional methods were chosen from among members of the Black Carrot Producer Union in Eregli, Konya. The stratified random sampling method was used in the selection of the 210 total population of the farms. The sampling equation (Yamane 1967) to determine sample size is:

$$n = \frac{N * s^2 * t^2}{(N-1)d^2 + s^2 * t^2} \quad (1)$$

where n is the required sample size, N is the number of holdings in target population, s is the standard deviation, d is the precision where (x–X) and t is the reliability coefficient (1.96 which represents the 95% reliability).

Data on energy consumption pattern and benefit-cost analysis for organic and conventional black carrot came from the 2007–2008 production year. Secondary data and energy equivalents for energy input and energy output levels were obtained from previous studies and related literature.

In this study, the inputs used for cultivation practices and outputs for organic and conventional black carrot were first determined. All inputs-outputs from the systems were identified, quantified and converted into energy units. The energy equivalents of the inputs and outputs were taken from a number of articles (Table 1). The black carrot seeds in the research area were produced by the farmers. Therefore, the seed energy equivalent was calculated for common production techniques. Energy consumption, on the other hand, was calculated by using conversion factors (1 L diesel= 56.31 MJ) and the value was expressed in MJ ha^{-1} (Ozkan et al. 2004). The energy input can also be classified into direct and indirect (Risoud 1999; Uhlin 1998), and renewable and non-renewable forms (Yilmaz et al. 2005). Indirect energy consists of seed, manure, pesticide, machinery and fertilizer while direct energy includes human power, diesel and electricity energy used in the production process. Non-renewable energy includes diesel, electricity, pesticide, machinery and fertilizers. Renewable energy consists of human power, seed and manure. Output-input ratio and energy efficiency of black carrot production were also calculated by using the following equations (Baishya and Sharma 1990; Burnett 1982; Shresta 1998):

Output-Input Ratio = Energy Output (MJ ha^{-1})/ Energy Input (MJ ha^{-1})

Table 1. Energy equivalents of inputs and outputs in agricultural production.

Input (Unit)	Energy Equivalent (MJ unit^{-1})
Human labor (h)	2.30
Machinery (h)	64.806
Chemical fertilizers (kg)	
Nitrogen fertilizer	66.14
Phosphorus	12.44
Potassium	11.15
Pesticide	
Herbicides	238.0
Fungicides	92.0
Manure (kg)	0.30
Seeds (kg)*	79,45
Diesel oil (L)	56.31
Electricity (kWh)	3.60
Irrigation water (m^3)	0.63
Output (kg)	1.96

Sources: Singh (2002); Singh et al. (2002); Yaldiz et al. (1993); Onal and Tozan (1986); Mandal et al. (2002); Titchenal and Dobbs (2004).
*Calculated in this research

Energy Productivity = Black Carrot Output (kg ha^{-1})/ Energy Input (MJ ha^{-1})

Net profit was obtained by subtracting the production cost of the conventional and organic black carrot per hectare from the gross production value. Cost/benefit ratios of the organic and conventional black carrot farming systems, on the other hand, were calculated by dividing gross production value (GPV) by the total production costs.

The Cobb-Douglas function was used in the study in order to estimate the effects on the production function of all inputs as well as the implemented farming system for black carrot.

The form of the Cobb-Douglas function for the study was:

$$Y = A \prod_{i=1}^{n} X_i^{\beta_i}, \qquad \beta_i \succ 0;\ i = 1, 2, \ldots, n.,$$

where Y is the output (Energy Output (MJ/ha), X represents a vector of essential inputs used in production, n is the total number and the parameter A is the combined effects on the production function of all inputs that are indicators for energy (Hatirli et al. 2006). Empirically, the impact of all inputs for conventional and organic black carrot farming on the amount of production becomes as follows:

$$\ln(Y) = \ln(A) + \sum_{i=1}^{n} \beta_i \ln(X_i) + \sum_{j=1}^{J} \delta_j D_j + \varepsilon;$$

$$\beta_i \succ 0,\ i = 1,2,\ldots,n;\ \ j = 1,2,\ldots,J.$$

where Y is energy output measured per unit land; A is the constant coefficient; X_i represents the variable inputs such as human labor, fuel, electricity, machinery, fertilizer, seed, pesticide, manure and water used per unit of land; D_j are dummy variables for farming systems (conventional or organic); β_i is the coefficient of inputs estimated from the model; δ_j is the coefficient of the energy differences obtain from organic and conventional carrot production; and ε is the error term of the regression equation.

RESULTS AND DISCUSSION

The most common management practices of conventional and organic black carrot farming systems in the region were taken as a base for each process in the research. The management practices of organic and conventional black carrot farming were generally similar except for the use of chemical fertilizer, herbicide and pesticide. Even though chemicals were not used in organic farming, there was more need of human power in the organic system than in the conventional system. The earliest implemented operations for both conventional and organic black carrot farming systems were soil tillage (performed from February to March) and seeding (March to April). The remaining practices were done from May through September. The harvest period, on the other hand, occurred from September to November. Harvesting was carried out by using traditional methods, i.e., carrots are removed from the soil by tractor and human labor. Transport was done by tractors pulled by trailer at an average distance of over 20 km. The inputs used according to the common management practices in conventional and organic black carrot production in the survey area and their energy equivalents with output energy rates are shown in Table 2.

The results revealed that 464.7 h of human labor and 84.8 h of machinery (tractor) power per hectare were required to conventionally produce black carrot in the research area. About 78.72% of the total human labor was used on cultural practices while the remaining labor was used on land preparation and harvesting. About 50% of the machine power in conventional black carrot growing

Table 2. Energy consumption and energy efficiency (energy input-output relationship) of conventional and organic black carrot farming systems.

Inputs (unit)	Total Energy Equivalent (MJ ha^{-1})	Conventional Black Carrot			Organic Black Carrot		
		Quantity Used per Unit Area (ha)	Total Energy Equivalent (MJ)	%	Quantity Used per Unit Area (ha)	Total Energy Equivalent (MJ)	%
Human Labor (h)		464.7	1,068.81	1.42	660.4	1,518.92	4.02
Land preparation		13.8	31.74		19.0	43.70	
Cultural practices	2.3	365.8	841.34		564.5	1,298.35	
Harvesting and transport		85.1	195.73		76.9	176.87	
Machinery (h)		84.8	5,316.96	7.06	71.8	4,501.86	11.92
Land preparation		42.4	2,658.48		35.9	2,250.93	
Cultural practices	62.7	8.9	558.03		8.7	545.49	
Harvesting and transport		33.5	2,100.45		27.2	1,705.44	
Chemical fertilizers (kg)		568.2	25,002.21	33.19	-	-	
Nitrogen (N)	66.14	343.1	22,692.63		-	-	
Phosphorus (P_2O_5)	12.44	140.8	1,751.55		-	-	
Potassium (K_2O_5)	11.15	49.1	547.47		-	-	
Farm yard manure	0.30	35.2	10.56		238.8	74.73	0.20
Pesticides (kg)		5.23	1,040.34	1.38	-	-	
Herbicides	238.0	3.83	911.54		-	-	
Fungicides	92.0	1.40	128.80		-	-	
Seeds (kg)	79.45	8.1	643.54	0.85	8.0	635.60	1.68
Diesel oil (L)	56.31	244.1	13,745.27	18.24	236.4	13,311.68	35.26
Electricity (kWh)	11.93	1,806.6	21,552.74	28.61	1,005.0	11,989.65	31.75
Irrigation water (m^3)	0.63	11,056.9	6,965.85	9.25	9,089.49	5,726.38	15.17
Total energy (MJ ha^{-1})			75,335.72	100.00		37,758.82	100.00
Yield (kg ha^{-1})	1.96	49,503.95	97,027.74		36,649.64	71,833.29	100.00
Energy output–input ratio			1.30			1.90	
Energy Productivity (kg MJ^{-1})			0.66			0.97	

was used for land preparation, followed by harvest and transport (39.50%), and cultural practices (10.50%). The amount of fertilizers used for conventional black carrot growing was 568.2 kg ha^{-1}. Nitrogen (N) was the most common chemical fertilizer used (60.38%), followed by phosphorus (P_2O_5) (24.78%), potassium (K_2O_5) (8.64%) and farm manure (6.20%). The total energy used for various processes in the conventional growing of black carrot was calculated to be 75,335.72 MJ ha^{-1}. Diesel oil, which was mainly utilized for operating tractors to perform the various farm operations, had the largest proportional share (35.26%) in the total energy used on organic black carrot, followed by electricity (31.75%), water for irrigation (15.17%) and machinery (11.92%). The energy input of human labor took a share of 4.02% in the total energy input use. The total amount of energy used in the various production processes for organic black carrot was calculated as 71,833.29 MJ ha^{-1}.

Yields of conventional and organic black carrot were calculated as 49,503.95 kg ha^{-1} and 36,649.64 kg ha^{-1},

respectively, in the survey area. Yield of organic black carrot was lower than that of conventionally produced carrot even though producers used organic manure. The calculated values of total energy output per hectare were 97,027.74 MJ ha^{-1} in conventional and 71,833.29 MJ ha^{-1} in organic black carrot production. Energy output/input ratios were determined as 1.30 and 1.90 in conventional and organic black carrot production, respectively. On the other hand, energy productivity was higher for organic black carrot production.

Many studies have been conducted on energy input-output issues and economic analysis for various crops. A comparative study was conducted in Canada for two crop rotations (wheat-pea-wheat-flax and wheat-alfalfa-alfalfa-flax) that were cultivated both organically and conventionally, and concluded that the energy used was 50% lower for organic farming than for conventional farming. Even though the energy output was lower for the organic system, the energy efficiency (energy produced /energy used) remains better in organic agriculture. Organic agriculture, with its reliance on natural fertilizers and chemicals, often performs relatively better in terms of energy efficiency (measured as the ratio of energy input per unit of crop output) despite lower yields.

An analysis of conventional rice farming in the Philippines concluded that rice yield increased by 6- to 25-fold with energy consumption primarily because of the fertilizer used to achieve incremental gains in crop yield (Pretty 1995). Nguyen and Haynes (1995) compared low-input conventional and organic integrated sheep and arable crop farms in New Zealand in terms of their energy efficiency. They found similar energy efficiency ratios for organic and conventional farms, with the organic ratios slightly better in two of the three pairs of comparisons, due to reduced reliance on synthetic fertilizers and chemicals in the conventional system. Organically produced tea in an integrated agroforestry system has also been found to save energy compared with its conventional counterpart by relying on biological fertilizers such as manure and crop residue instead of synthetics (Jianbo 2006).

Total energy input consumption was classified into direct energy (57.52%), indirect energy (42.48%), renewable energy (11.51%) and non-renewable energy (88.49%) in conventional black carrot production (Table 3). Nonetheless, as also shown in the table, total energy used in organic black carrot farming system consisted mainly of direct energy (86.20%), non-renewable energy (78.93%), indirect energy (13.80%) and renewable energy (21.07%).

Cost/benefit analysis was performed by taking variable and fixed cost components into consideration, including production costs and gross value for conventional and organic carrot farming systems, separately (Table 4).

Total production cost of conventional and organic black carrot were 5,421.27 \$ ha^{-1} and 5,005.56 \$ ha^{-1}, respectively, while gross production values (GPV) were 9,900.79 \$ ha^{-1} and 10,261.0 \$ ha^{-1}, respectively. The total production costs consisted of 66.17% and 69.67% of variable costs while 33.83% and 30.33% of these costs were fixed costs, respectively, for conventional and organic black carrot farming systems. The highest production cost was spent on wages paid to casual workers, followed by land rent and repair-maintenance expenses. The production cost per hectare on organic farms was 7.7% less than it was on conventional farms. Gross production value (GPV) was conversely 3–5% higher than the value for conventional farms. This result was due to the sales prices, with organic black carrot prices 28.6% higher than those of conventional black carrot. Thus, the net profit of organic black carrot was higher (5,256.34 \$ ha^{-1}) than that of conventional black carrot (4,479.52 \$ ha^{-1}). Cost/benefit ratios of conventional and organic black carrot farming systems were found to be 1.83 \$ ha^{-1} and 2.05 \$ ha^{-1}, respectively, with the ratio higher for the organic farms compared with the conventional farms.

The coefficient for the farming system was positive and significantly different with a level of 5%. This means that, for the same levels of all inputs, only the employment of organic farming increased the amount of the energy output compared with the conventional system. Most other inputs, i. e., human labor, machinery, fertilizer, fuel (diesel oil) and seeds, significantly contributed to the increase in the energy output for black carrot production. Electricity, pesticide, manure and irrigation water were not significant. The Adj R-squared (R^2) equals 92.4 in black carrot production.

The organic system was statistically different, with a level of 1%, in terms of energy output. This analysis indicates that 1 unit (man power) of human labor increases energy output by 12.53 kg ha^{-1}. The use of 1 kg of fertilizer increases energy output by approximately 2.9 kg ha^{-1}. The organic system may increase energy output by 1.32 kg ha^{-1} whereas the energy increase becomes 0.23 kg ha^{-1} for the conventional system.

CONCLUSION

Increasing amount of inputs was more effectively used in the organic farming system compared with the conventional system. Total energy requirement under conventional farming was 75,335.72 MJ ha^{-1} and

Table 3. Total energy input in the form of direct, indirect, renewable and non-renewable for conventional and organic carrot farming systems (MJ ha^{-1})

Type of Energy	Conventional Black Carrot	%	Organic Black Carrot	%
Direct energy[a]	43,332.67	57.52	32,546.69	86.20
Indirect energy[b]	32,003.05	42.48	5,212.13	13.80
Renewable energy[c]	8,668.76	11.51	7,955.63	21.07
Non-renewable energy[d]	66,666.96	88.49	29,803.19	78.93
Total energy input	75,335.72	100.00	37,758.82	100.00

[a] Includes human, diesel and electricity, water for irrigation energy sources.
[b] Includes seeds, fertilizers, manure, chemicals and machinery energy sources.
[c] Includes human, seeds, manure and water for irrigation.
[d] Includes diesel, electricity, pesticides, fertilizers and machinery

Table 4. Cost-benefit analyses of conventional and organic black carrot farming systems.

	Conventional Black Carrot		Organic Black Carrot	
Costs	**($ ha^{-1})**	**(%)**	**($ ha^{-1})**	**(%)**
Variable Costs	**3,587.14**	**66.17**	**3,487.62**	**69.67**
Seeds	250.48	4.62	244.68	4.89
Fertilizers	296.75	5.47	81.75	1.63
Fungicides	65.48	1.21	0.00	0.00
Herbicides	38.89	0.72	0.00	0.00
Casual workers	918.33	16.94	1,655.16	33.07
Water	62.54	1.15	59.76	1.19
Diesel	430.56	7.94	412.78	8.25
Electricity	172.06	3.17	113.65	2.27
Repair and maintenance	130.87	2.41	611.90	12.22
Others	26.83	0.49	34.68	0.69
Operating interest charges	265.79	4.90	273.25	5.46
Fixed Costs	**1,834.13**	**33.83**	**1,517.94**	**30.33**
Family labor	204.21	3.77	217.06	4.34
Depreciation	165.00	3.04	140.87	2.81
Interest	354.37	6.54	163.57	3.27
Land rent	1,003.41	18.51	891.83	17.82
General overhead cost (%)	107.62	1.99	104.60	2.09
Total Production Costs	**5,421.27**	**100.00**	**5,005.56**	**100.00**
Gross Production Value (GPV)	9,900.79	-	10,261.90	-
Net Profit	4,479.52	-	5,256.34	-
Benefit/Cost ratio	1.83	-	2.05	-
Productivity (kg $\$^{-1}$)	9.13	-	7.32	

[a]Total production value = black carrot yield (kg ha^{-1}) multiplied by black carrot price ($ kg^{-1}).
[b]Productivity (kg $\$^{-1}$) = black carrot yield (kg ha^{-1}) divided by total production costs ($ ha^{-1}).
(1US$ = YTL 1.26 in production period).

Table 5. Estimated parameter for energy output using the Cobb-Douglas function.

Variable	Coefficient	t-statistic
Human labor	1.253***	14.540
Machinery	0.354*	3. 920
Fertilizer	0.290*	2.250
Seeds	0.054***	11.247
Fuel (diesel oil)	0.520**	3.22
Electricity	0.009	0.94
Pesticide	0.038	0.02
Manure	0.004	0.07
Irrigation water	0.007	0.82
Organic system	0.132**	2.871
Conventional system	0.023*	1.212

Adj R-squared equals 92.4.
*, ** and *** indicate significance at 10%, 5% and 1% levels, respectively.

37,758.82 MJ ha^{-1} for organic farming. There was higher demand for energy in conventional farming (49.88%) than organic farming. On the other hand, the need for labor per hectare was higher for organic farming than for the conventional farming system. The energy output-input ratios for conventional and organic farming systems were 1.30 and 1.90, respectively. Moreover, the benefit-cost ratios were 1.83 and 2.05, respectively. Energy efficiency in the organic systems may increase by 1.32 kg ha^{-1} and by 0.23 kg ha^{-1} for the conventional systems.

Energy efficiency (energy input-output analysis) is closely associated with economic (profitability) and ecological aspects of the chosen farming systems. Energy efficiency and energy balance can be accepted as a vital tool to determine the environmental impacts of farming systems. Determination of the energy efficiency makes it possible to compare different farming systems in environment friendly production as well as sustainability of non-renewable natural resources. Three of the 5Es – Energy efficiency, Economic profitability and Environment friendly production – of the sustainable farming systems were observed to be interacting with one another. Farmers' production practices (Experiences) and food safety (Ethics of a profession), as determinative factors for integrated farming system management (IFSM), can be taken into consideration to compare the advantages, with factors such as processing and marketing opportunities, for the long-term farming objectives. It is recommended that more research be done on IFSM with respect to technological developments, farmers' experiences, environment friendly production and economic issues (specifically the agribusiness component).

REFERENCES CITED

ANONYMOUS. 2007a. Records of Konya State Meteorology Station. Konya, Turkey.

ANONYMOUS. 2007b. Ministry of Agriculture and Rural Affair, Konya branch records, Turkey.

BAISHYA A, SHARMA GL. 1990. Energy budgeting of rice-wheat cropping system. Indian J Agron 35(1,2):167-177.

BURNETT M. 1982. Energy analysis of three agro-eco systems, In: Hill SB, editor. Paper presented at the Second International Conference on Basic Techniques in Ecological Farming; 1-5 October; IFO, Montreal, Basel: Birkhauser. p. 183-195.

EDWARDS CA. 1987. The concept of integrated systems in lower input/sustainable agriculture. Am J Alternative Agric 2:148-152.

GUNDOGMUS E. 2006. Energy use on organic farming: A comparative analysis on organic versus conventional apricot production on small holdings in Turkey. Energy Conversion and Management 47:3351-3359.

GÜNDOGMUS E, BAYRAMOGLU Z. 2005. Energy input use on organic farming: A comparative analysis on organic versus conventional farms in Turkey. J Agron 5(1):16-22.

HATIRLI SA, OZKAN B, FERT C. 2006. Energy inputs and crop yield relationship in greenhouse tomato production. Renewable Energy 31:427-438.

JIANBO L. 2006. Energy balance and economics benefits of two agroforesty systems in Northern and Southern China. Agriculture, Ecosystems and Environment 116 (3-4): 255-262.

KLIMEKOVA M, LEHOCKA Z. 2007. Comparison of organic and conventional farming system in term of energy efficiency. Poster presented at Zwischen Tradition und Globalisierung–9. Wissenschaftstagung Ökologischer Landbau, Universität Hohenheim, Deutschland.

MÄDER P, FLIEBBACH PA, DUBOIS D, GUNST L, FRIED P, NIGGLI U. 2002. Soil Fertility and Biodiversity in Organic Farming. Science 296:1694-1697.

MANDAL KG, SAHA KP, GOSH PL, HATI KM,

BANDYOPADHYAY KK. 2002. Bioenergy and economic analyses of soybean-based crop production systems in Central India. Biomass Bioenergy 23:337-345.

NGUYEN ML, HAYNES RJ. 1995. Energy and labour efficiency for three pairs of conventional and alternative mixed cropping (pasture-arable) farms in Canterbury, New Zealand. Agriculture, Ecosystems and Environment 52:163-172.

OGUZ C, PEKER K, OGUT H. 2005. The comparative advantages of biodiesel production from some oil crops to fossil fuels in Turkey, ISET2005. Proceedings of the 1st International Symposium and Exhibition on Environment-Friendly Energy Sources and Technologies; 5–7 September 2005. Altinyunus-Cesme, Turkey.

ONAL I, TOZAN M. 1986. Energy budget and work requirements of the alternative production systems in the processing tomato productions. In: 10th National Agricultural Mechanization Congress. Adana, Turkey, 5–7 May 1986. p. 216-28.

OZKAN B, AKCAOZ H, KARADENIZ F. 2004. Energy requirement and economic analysis of citrus production in Turkey. Energy Converts Manage 45:1821-1830.

PERVANCHON F, BOCKSTALLER C, GIRARDIN P. 2002. Assessment of energy use in arable farming systems by means of an agro-ecological indicator. Agric Syst 72: 149-172.

PIMENTEL D, HEPPERLY P, HANSON J, DOUDS D, SEIDEL R. 2005. Environmental, energetic, and economic comparison of organic and conventional farming systems. BioScience 55(7):573-582.

PRETTY J. 1995. Regenerating Agriculture: Policies and Practice for Sustainability and Self-reliance. London: Earthscan Publications Limited.

RISOUD B. 1999. Development durable et analyses energy etique de exploitations agricoles. Econ Rur 252:16-27.

SAYIN C, MENCET MN, OZKAN B. 2005. Assessing of energy policies based on Turkish agriculture: current status and some implications. Energy Policy 33:2361-2373.

SHRESTA DS. 1998. Energy use efficiency indicator for agriculture, 1998. http://www.usaskca/agriculture/caedac/PDF/mcrae.PDF,10/10/2002.

SINGH JM. 2002. Energy use pattern in different cropping systems in Haryana, India, [Master of Science thesis]. Germany: Institute of Management, University of Flensburg.

SINGH H, MISHRA D, NAHAR NM. 2002. Energy use pattern in production agriculture of a typical village in arid zone India. Energy Conversion and Management 43:2275-2286.

TITCHENAL AC, DOBBS J. 2004. Nutritional value of vegetables. In: Hui YH, Ghazala S, Graham DM, Murrell KD, Nip W, editors. Handbook of Vegetable and Preservation Processing. New York: Marcel Dekker.

TIWARI DN, LOOF R, PAUDYAL DN. 1999. Environmental-economic decision making in lowland irrigated agriculture using multi-criteria analysis techniques. Agric Syst 60(2):99-112.

TURYEREEBA PJ. 2001. Renewable energy, its contribution to improved standards of living and modernization of agriculture in Uganda. Renewable Energy 24:453-457.

UHLIN H. 1998. Why energy productivity is increasing: an I–O analysis of Swedish agriculture. Agric Syst 56(4):443-465.

VEREIJKEN P. 1997. A methodical way of prototyping integrated and ecological arable farming systems (I/EAFS) in interaction with pilot farms. Eur J Agron 7:235-250.

VLECK P, RODRIGUEZ-KUHL G, SOMMER R. 2004. Energy use and CO_2 production in tropical agriculture and means and strategis for reduction or mitigation. Bonn, Germany: Zentrum f˙ur Entwicklungsforschung (ZEF), Walter–Flex–Straße 3, 53113.

YALDIZ O, OZTURK HH, ZEREN Y, BASCETINCELIK A. 1993. Energy usage in production of field crops in Turkey. 5th International Congress on Mechanization and Energy Use in Agriculture, 11–14 October 1993; Izmir, Turkey.

YAMANE T. 1967. Elementary Sampling Theory. Englewood Cliffs, New Jersey, U.S.A.: Prentice Inc.

YILMAZ I, AKCAOZ H, OZKAN B. 2005. An analysis of energy use and input-output costs for cotton production in Turkey. Renew Energy 30:145-155.